SKILL EXPERIMENT - 9

Global Exception Handling using @ControllerAdvice

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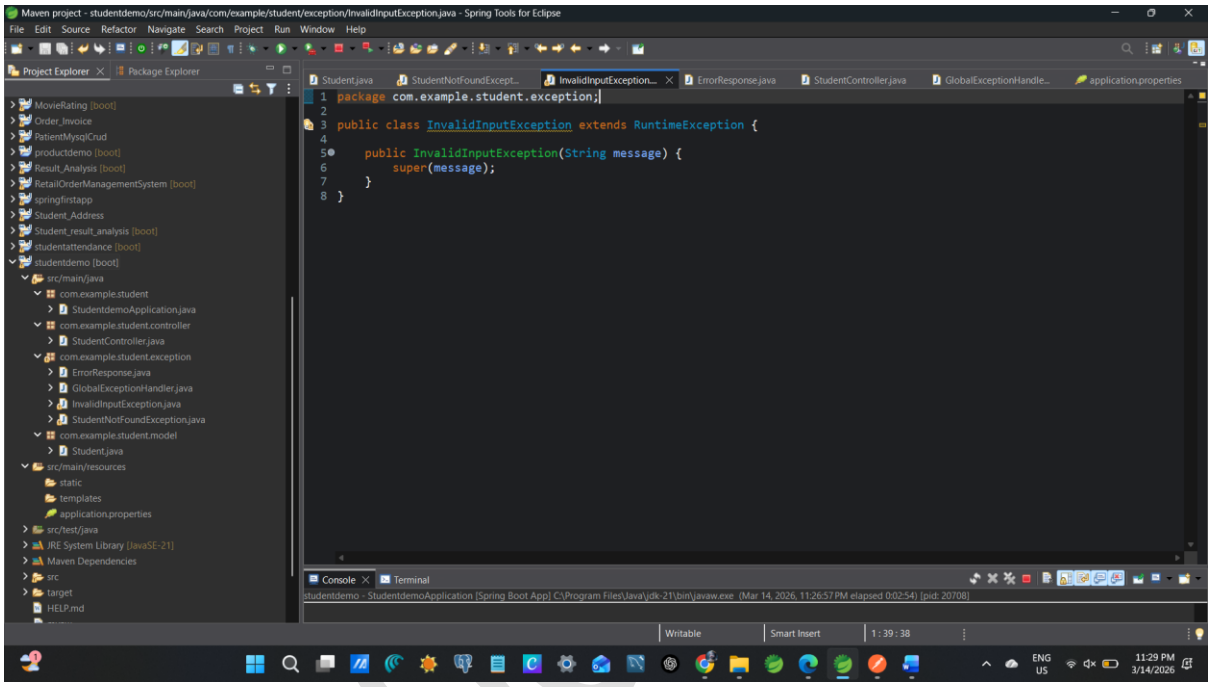
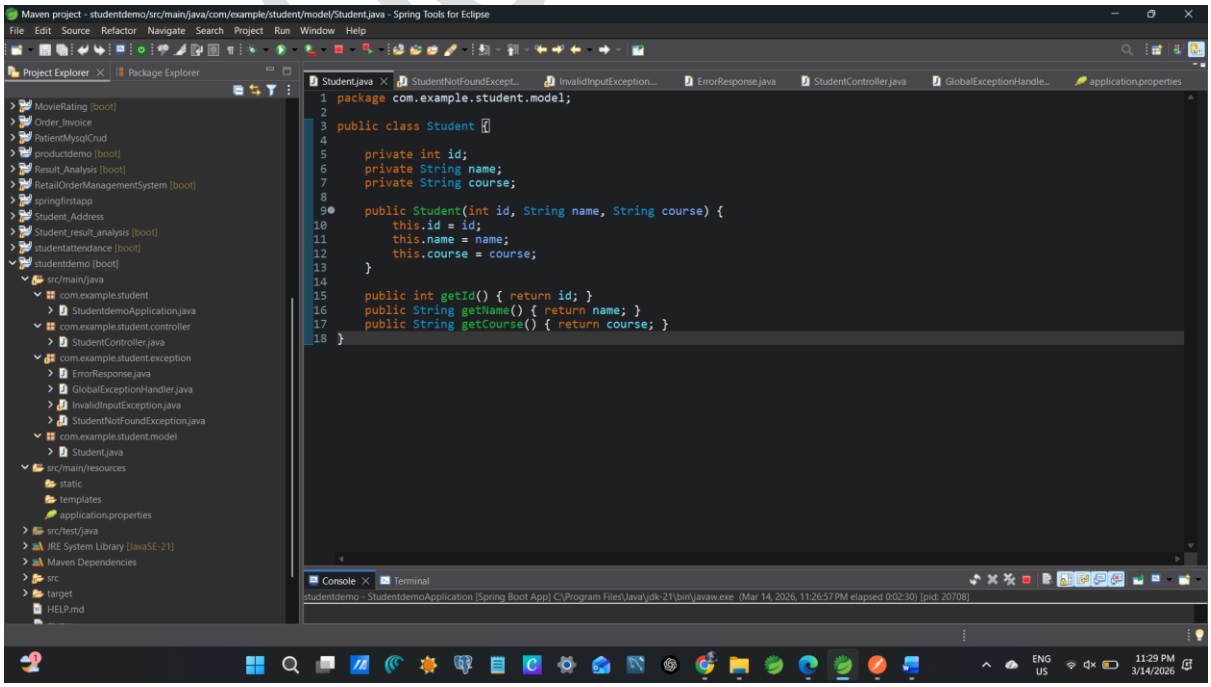
FSAD, Sec 4





Maven project - studentdemo/src/main/java/com/example/student/exception/StudentNotFoundException.java - Spring Tools for Eclipse

```java
package com.example.student.exception;

public class StudentNotFoundException extends RuntimeException {

    public StudentNotFoundException(String message) {
        super(message);
    }
}
```

Maven project - studentdemo/src/main/java/com/example/student/exception/ErrorResponse.java - Spring Tools for Eclipse

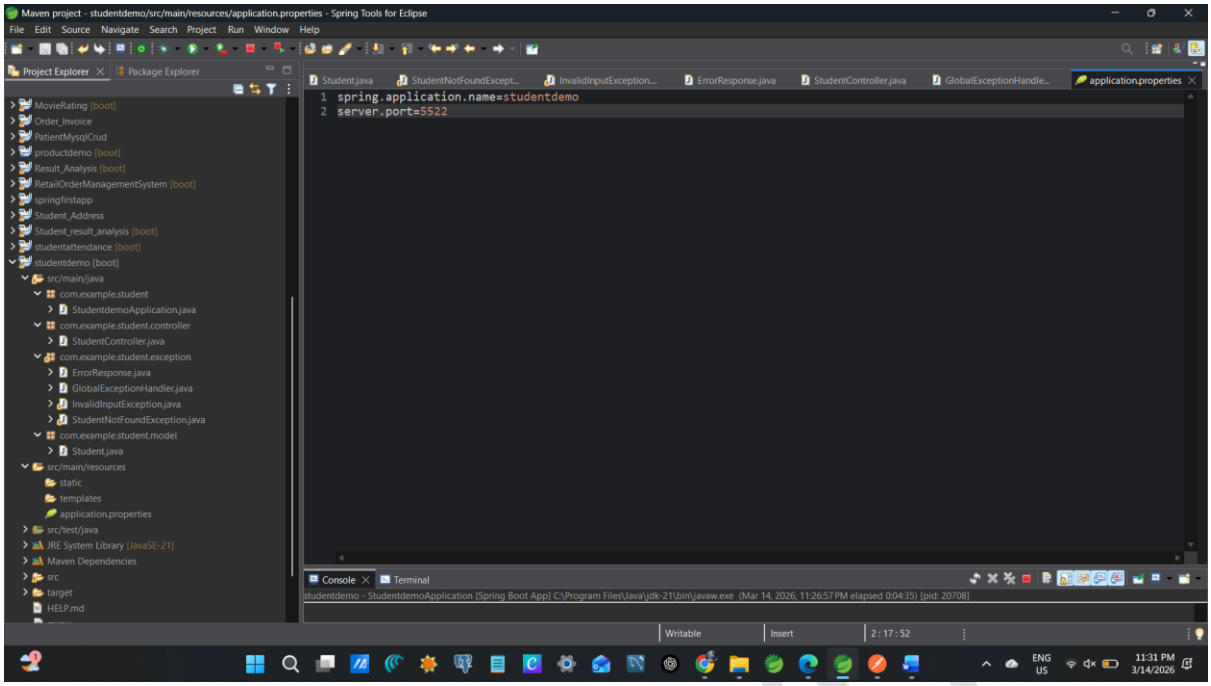
```java
package com.example.student.exception;

import java.time.LocalDateTime;

public class ErrorResponse {

    private LocalDateTime timestamp;
    private String message;
    private int statusCode;

    public ErrorResponse(LocalDateTime timestamp, String message, int statusCode) {
        this.timestamp = timestamp;
        this.message = message;
        this.statusCode = statusCode;
    }

    public LocalDateTime getTimestamp() { return timestamp; }
    public String getMessage() { return message; }
    public int getStatusCode() { return statusCode; }
}
```

Maven project - studentdemo/src/main/java/com/example/student/controller/StudentController.java - Spring Tools for Eclipse

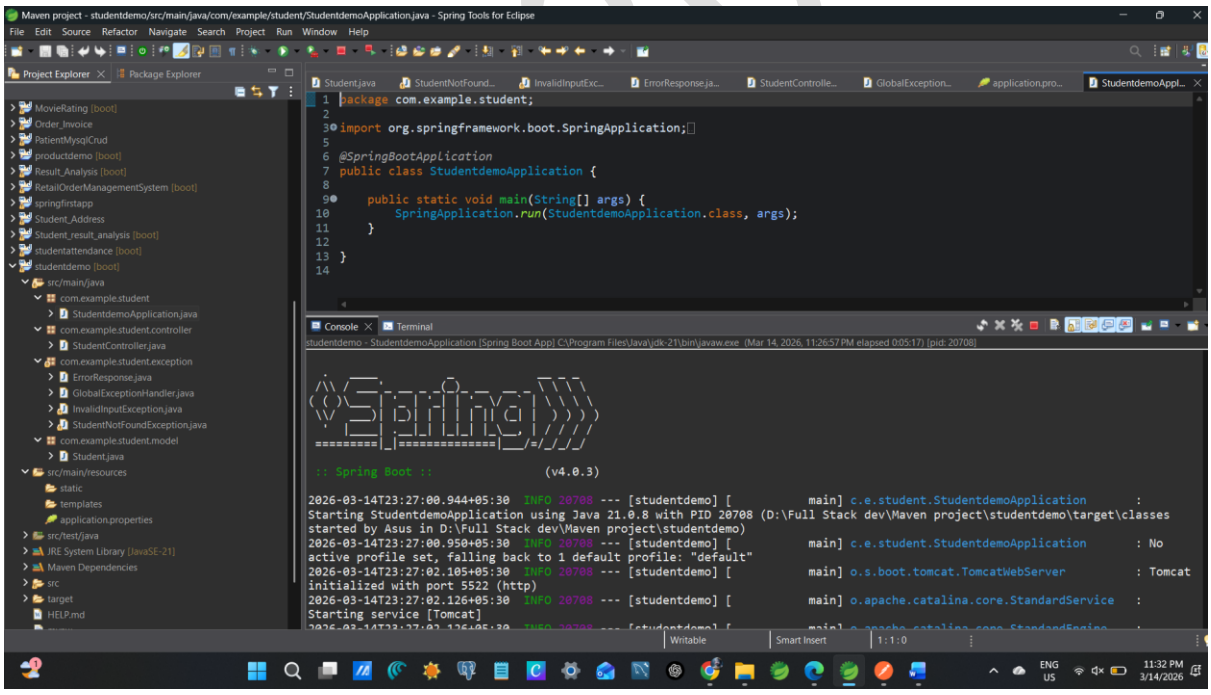
```java
package com.example.student.controller;

import com.example.student.model.Student;
import com.example.student.exception.StudentNotFoundException;
import org.springframework.web.bind.annotation.*;

@RestController
public class StudentController {

    @GetMapping("/student/{id}")
    public Student getStudent(@PathVariable int id) {

        if(id != 1){
            throw new StudentNotFoundException("Student with ID " + id + " not found");
        }

        return new Student(1,"Rahul","CSE");
    }
}
```

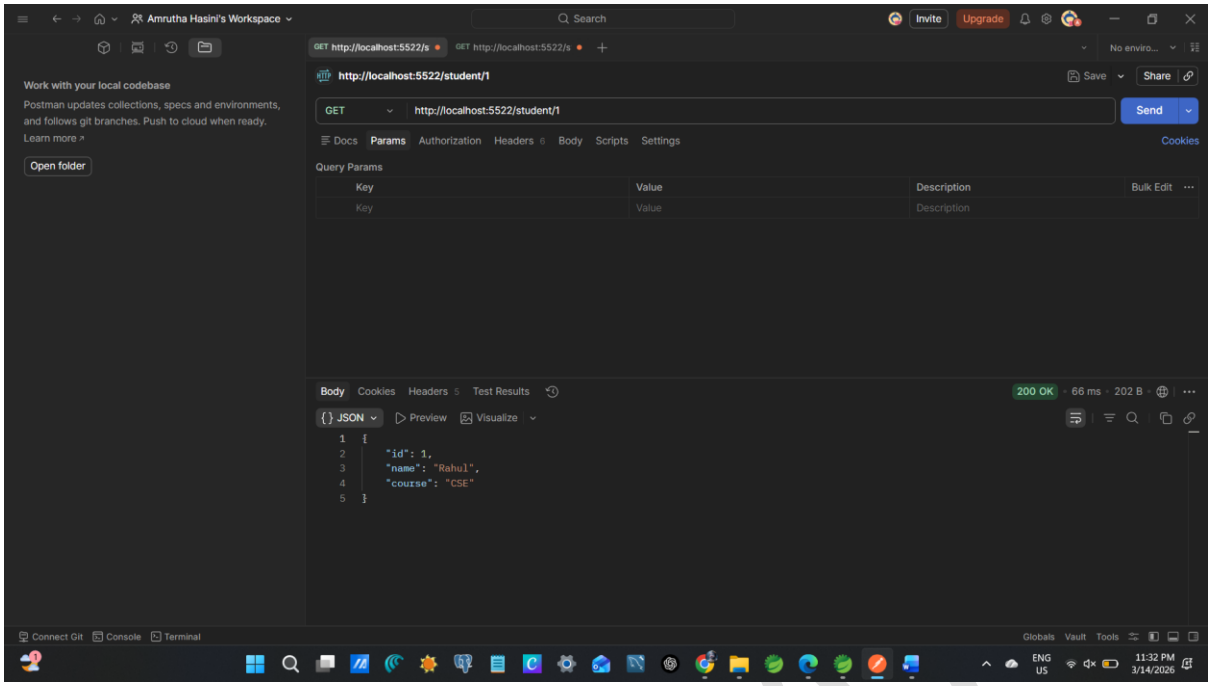
Maven project - studentdemo/src/main/java/com/example/student/exception/GlobalExceptionHandler.java - Spring Tools for Eclipse

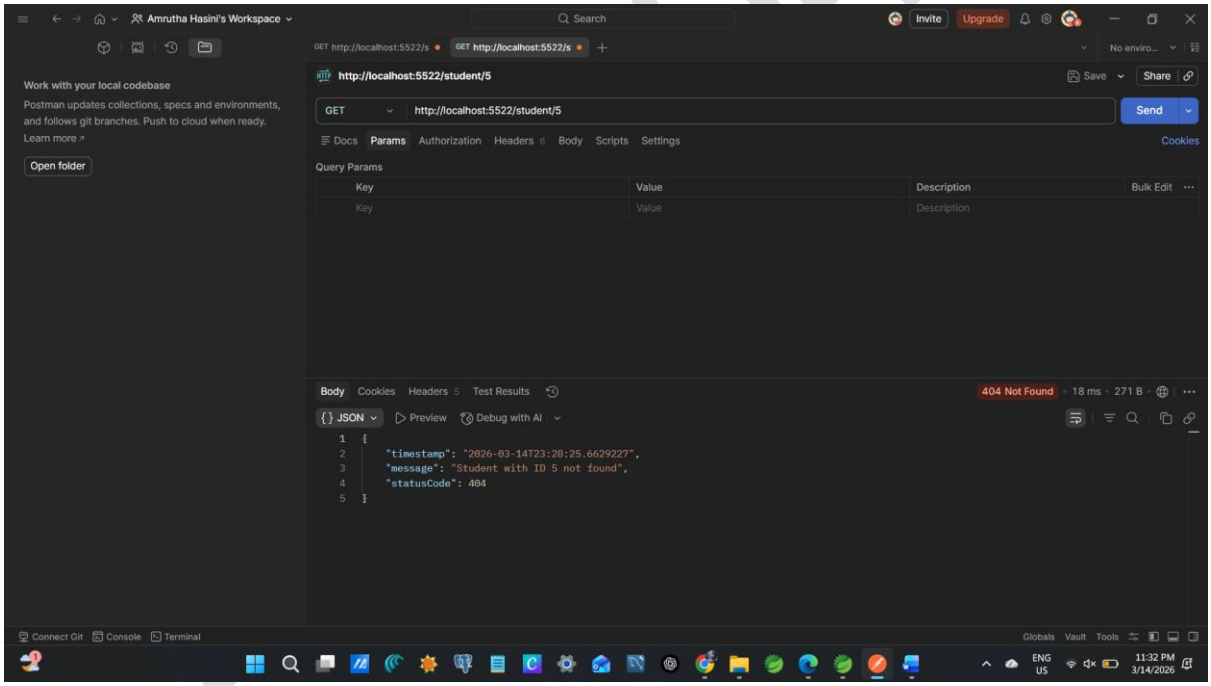
```java
package com.example.student.exception;

import org.springframework.http.*;
import org.springframework.web.bind.annotation.*;

import java.time.LocalDateTime;

@ControllerAdvice
public class GlobalExceptionHandler {

    @ExceptionHandler(StudentNotFoundException.class)
    public ResponseEntity<ErrorResponse> handleStudentNotFound(StudentNotFoundException ex){

        ErrorResponse error = new ErrorResponse(
                LocalDateTime.now(),
                ex.getMessage(),
                HttpStatus.NOT_FOUND.value()
        );

        return new ResponseEntity<>(error,HttpStatus.NOT_FOUND);
    }

    @ExceptionHandler(InvalidInputException.class)
    public ResponseEntity<ErrorResponse> handleInvalidInput(InvalidInputException ex){

        ErrorResponse error = new ErrorResponse(
                LocalDateTime.now(),
                ex.getMessage(),
                HttpStatus.BAD_REQUEST.value()
        );

        return new ResponseEntity<>(error,HttpStatus.BAD_REQUEST);
    }
```


Maven project - studentdemo/src/main/resources/application.properties - Spring Tools for Eclipse
File Edit Source Navigate Search Project Run Window Help
Project Explorer
Package Explorer
Student.java
StudentNotFoundExcept...
InvalidInputException...
ErrorResponse.java
StudentController.java
GlobalExceptionHandle...
application.properties
1 spring.application.name=studentdemo
2 server.port=5522
Console
Terminal
studentdemo - StudentdemoApplication [Spring Boot App] C:\Program Files\Java\jdk-21\bin\javaw.exe (Mar 14, 2026, 11:26:57 PM elapsed 0:04:35) [pid: 20708]
Writable
Insert
2 : 17 : 52
ENG
US
11:31 PM
3/14/2026

OUTPUTS: -


Maven project - studentdemo/src/main/java/com/example/student/StudentdemoApplication.java - Spring Tools for Eclipse
File Edit Source Refactor Navigate Search Project Run Window Help
Project Explorer
Package Explorer
Student.java
StudentNotFound...
InvalidInputExc...
ErrorResponse.ja...
StudentControlle...
GlobalException...
application.pro...
StudentdemoAppl...
1 package com.example.student;
3 import org.springframework.boot.SpringApplication;
6 @SpringBootApplication
7 public class StudentdemoApplication {
9 public static void main(String[] args) {
10 SpringApplication.run(StudentdemoApplication.class, args);
11 }
13 }
Console
Terminal
studentdemo - StudentdemoApplication [Spring Boot App] C:\Program Files\Java\jdk-21\bin\javaw.exe (Mar 14, 2026, 11:26:57 PM elapsed 0:05:17) [pid: 20708]
:: Spring Boot :: (v4.0.3)
2026-03-14T23:27:00.944+05:30 INFO 20708 --- [studentdemo] [main] c.e.student.StudentdemoApplication :
Starting StudentdemoApplication using Java 21.0.8 with PID 20708 (D:\Full Stack dev\Maven project\studentdemo\target\classes
started by Asus in D:\Full Stack dev\Maven project\studentdemo)
2026-03-14T23:27:00.950+05:30 INFO 20708 --- [studentdemo] [main] c.e.student.StudentdemoApplication : No
active profile set, falling back to 1 default profile: "default"
2026-03-14T23:27:02.105+05:30 INFO 20708 --- [studentdemo] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat
initialized with port 5522 (http)
2026-03-14T23:27:02.126+05:30 INFO 20708 --- [studentdemo] [main] o.apache.catalina.core.StandardService :
Starting service [Tomcat]
Writable
Smart Insert
1 : 1 : 0
ENG
US
11:32 PM
3/14/2026





GitHub repo: - https://github.com/AmruthaHasini-KLU/skill-9.git